Course support and deadlines

Ed Discussion: questions, announcements, and private messages. No Canvas Inbox.

Open Ed from Canvas All times Pacific Full contact details and meeting links

Instructor / TAs	Office hours	Optional, ungraded discussions
Antonio de Leon GSC (in person)	**Mon 3–4 p.m. (extra)** Fri 3–4 p.m.	—
Jeannine Romero	Tue / Thu 10–11 a.m.	01A: Fri 9:20–10:25 a.m. R Carson Acad 240 (in person)
Ehsan Mirafzali	Mon 3–5 p.m.	01B: Fri 10:40–11:45 a.m. (Zoom)
Seokjun Choi	Wed 3–4 p.m. 5:05–6:05 p.m.	01C: Wed 4–5:05 p.m. (Zoom)

Upcoming deadlines HW and quizzes: **10 p.m. Pacific**

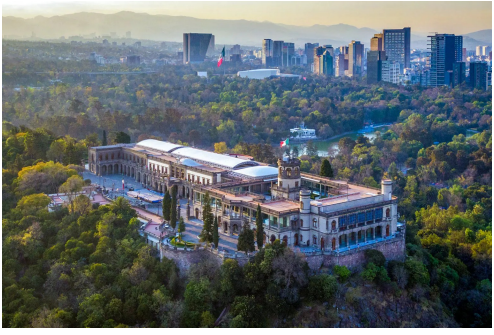
Quiz 4	**Sun, Oct 25**	Canvas
HW 3	**Tue, Oct 27**	Gradescope

GSC: Graduate Student Commons. TA office-hour times link to Zoom.

HW: 24-hour grace period. Quizzes: no grace period. Approved accommodations apply.



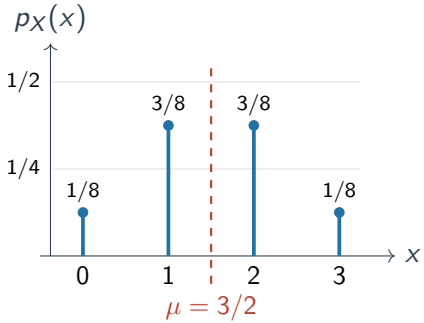




Same mean, different spread

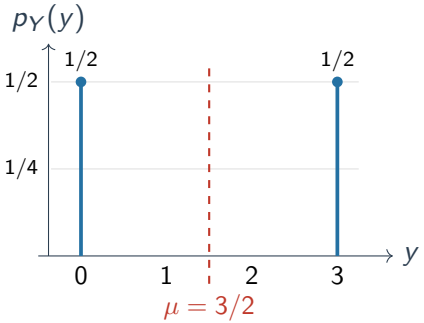
Three independent fair tosses

X counts heads: $X \sim \text{Bin}(3, 1/2)$.



A hypothetical all-or-none count

$Y = 0$ or 3, each with probability $1/2$.



Both means are $3/2$. Which law puts more probability far from its mean?

These are two separate count models; no joint relationship is assumed.

Expectation of a function

Finite support S_X and a real-valued function g

$$\mathbb{E}[g(X)] = \sum_{x \in S_X} g(x) p_X(x).$$

Derivation. Put $Z = g(X)$. Group all inputs x giving the same output z, then substitute into $\mathbb{E}[Z] = \sum_z z\, P(Z = z)$.

Evaluate g at each possible value, then average with the original weights.

We prove the finite case. For countable support, require $\sum_{x \in S_X} |g(x)| p_X(x) < \infty$.

First and second moments

For $X \sim \mathrm{Bin}(3, 1/2)$, with $S_X = \{0, 1, 2, 3\}$, complete the two weighted sums.

x	0	1	2	3	Sum
$p_X(x)$	1/8	3/8	3/8	1/8	1
$xp_X(x)$					
$x^2 p_X(x)$					

$$\mathbb{E}[X^2] = \sum_{x \in S_X} x^2 p_X(x), \qquad (\mathbb{E}[X])^2 = \Big(\sum_{x \in S_X} x p_X(x) \Big)^2.$$

The second moment averages the squared values.

Variance and standard deviation

Assume $\mathbb{E}[X^2] < \infty$ and write $\mu = \mathbb{E}[X]$.

Expected squared deviation

$$\sigma^2 = \mathrm{Var}(X) = \mathbb{E}[(X-\mu)^2], \qquad \mathrm{SD}(X) = \sigma = \sqrt{\mathrm{Var}(X)}.$$

Why square? Signed deviations cancel: $\mathbb{E}[X-\mu] = 0$.

Finite-PMF argument. Explain nonnegativity and the zero-variance case:

$$\mathrm{Var}(X) = \sum_{x \in S_X} (x-\mu)^2 p_X(x).$$

If X is measured in milliseconds, variance has units ms^2 and SD has units ms.

Computing variance from two moments

For $\mathbb{E}[X^2] < \infty$

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

Derive it from the definition. Keep $\mu = \mathbb{E}[X]$ fixed as a number.

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mu)^2].$$

Expand the square, use linearity, and substitute $\mathbb{E}[X] = \mu$.

Linearity applies to the expectation in the definition. Independence is not used.

Variance of the coin count

For $X \sim \text{Bin}(3, 1/2)$, we found $\mathbb{E}[X] = 3/2$ and $\mathbb{E}[X^2] = 3$.

Using centered values

$$\text{Var}(X) = \sum_{x=0}^{3} (x - 3/2)^2 p_X(x).$$

Using two moments

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

The methods agree. Equal means do not imply equal variances.

Shifts and changes of scale

Let $Z = aX + b$, with real constants a, b and $\mathbb{E}[X^2] < \infty$.

Affine change

$$\mathrm{Var}(aX + b) = a^2 \,\mathrm{Var}(X), \qquad \mathrm{SD}(aX + b) = |a| \,\mathrm{SD}(X).$$

Center first. With $\mu = \mathbb{E}[X]$, linearity gives $\mathbb{E}[Z] = a\mu + b$.

A shift leaves centered deviations unchanged; scaling multiplies their squares by a^2.

If $a = 0$, then $Z = b$ is constant and its variance is zero.

A delay measured in milliseconds

Hypothetical model. Each of three independent subtasks needs an extra operation with probability $1/2$. Let X count these extra operations.

$$L = 100 + 20X \text{ ms}, \qquad X \sim \text{Bin}(3, 1/2).$$

Use $\mathbb{E}[X] = 3/2$ and $\text{Var}(X) = 3/4$.

Quantity	Calculation	Units
$\mathbb{E}[L]$		ms
$\text{Var}(L)$		ms^2
$\text{SD}(L)$		ms

Check the units as well as the number.

Bernoulli variance

Let $I \sim \text{Bern}(p)$, where $0 \leq p \leq 1$.

$$P(I = 1) = p, \qquad P(I = 0) = 1 - p, \qquad \mathbb{E}[I] = p.$$

Use the support. Since $I \in \{0, 1\}$, we have $I^2 = I$.

One trial

$$\mathbb{E}[I] = p, \qquad \text{Var}(I) = p(1 - p), \qquad \text{SD}(I) = \sqrt{p(1 - p)}.$$

Compute and rescale a variance

w	0	1	3
$p_W(w)$	1/2	1/4	1/4

Find $\mathbb{E}[W]$, $\mathbb{E}[W^2]$, $\mathrm{Var}(W)$, and $\mathrm{Var}(5-2W)$.

For SD, the scale factor is $|-2| = 2$.

Variance summary

Meaning	$\mathrm{Var}(X) = \mathbb{E}[(X-\mu)^2]$
Computation	$\mathrm{Var}(X) = \mathbb{E}[X^2] - \mu^2$
Affine change	$\mathrm{Var}(aX+b) = a^2\,\mathrm{Var}(X)$
Standard deviation	$\mathrm{SD}(aX+b) = \lvert a\rvert\,\mathrm{SD}(X)$

Here $\mu = \mathbb{E}[X]$ and $\mathbb{E}[X^2] < \infty$.

Variance is not linear

For $I \sim \mathrm{Bern}(1/2)$,

$$\mathrm{Var}(2I) = 1 \qquad \text{but} \qquad 2\,\mathrm{Var}(I) = \frac{1}{2}.$$

Draw two balls uniformly **without replacement** from three red and two blue balls.



$$X = I_{\{\text{first draw is red}\}}, \qquad Y = I_{\{\text{second draw is red}\}}.$$

Each indicator equals 1 for red and 0 for blue. Here X and Y describe the two draws.

Physical outcome	Color pair	(X, Y)
(R_2, B_1)	red, blue	$(1, 0)$
(B_2, R_3)	blue, red	$(0, 1)$

The two variables describe the same outcome. Independence has not been assumed.

Joint probability mass functions

Discrete X and Y

$$p_{X,Y}(x,y) = P(X = x, Y = y).$$

The events $\{X = x, Y = y\}$ form disjoint cells covering the sample space.

$$p_{X,Y}(x,y) \geq 0, \qquad \sum_x \sum_y p_{X,Y}(x,y) = 1.$$

Justify normalization by adding the probabilities of the disjoint cells.

The joint support is the set of pairs with positive mass; put zero outside it.

All worked tables are finite. These nonnegative-sum formulas also hold for countable supports.

Two draws without replacement

There are $5 \cdot 4 = 20$ equally likely ordered pairs of distinct physical balls.

Count the pairs in each cell

$X \backslash Y$	0 (blue)	1 (red)
0 (blue)	$2 \cdot 1$	$2 \cdot 3$
1 (red)	$3 \cdot 2$	$3 \cdot 2$

The four counts are $2, 6, 6, 6$.

Normalize by the total count

$X \backslash Y$	0	1
0		
1		

Equal probability applies to physical-ball pairs, not automatically to color pairs.

Events in a joint table

Add the masses of the cells in the event:

$$P((X,Y) \in A) = \sum_{(x,y)\in A} p_{X,Y}(x,y).$$

The colors agree: $X = Y$

$X \backslash Y$	0	1
0	1/10	3/10
1	3/10	3/10

At least one red: $X + Y \geq 1$

$X \backslash Y$	0	1
0	1/10	3/10
1	3/10	3/10

Translate the event into cells before adding their masses.

Deriving a marginal PMF

Fix x. The possible values of Y split the event $\{X = x\}$:

$$\{X = x\} = \bigcup_y \{X = x, Y = y\} \qquad \text{(disjoint union)}.$$

Marginal distributions

$$p_X(x) = \sum_y p_{X,Y}(x, y), \qquad p_Y(y) = \sum_x p_{X,Y}(x, y).$$

Keep the variable you want; sum over the variable you remove.

Row totals and column totals

Complete the margins of the two-draw table.

$X \backslash Y$	0	1	$p_X(x)$
0	1/10	3/10	
1	3/10	3/10	
$p_Y(y)$			1

Check: each marginal has nonnegative entries summing to one.

Equal marginal distributions do not establish independence.

The total number of red balls

Let $N = X + Y$. Group cells by their total, then recover $\mathbb{E}[N]$ in two ways.

n	Pairs (x, y)	$p_N(n)$
0	$(0,0)$	$1/10$
1	$(0,1), (1,0)$	
2	$(1,1)$	$3/10$

Linearity gives the same answer even though these draws are dependent.

The same marginals, two joint laws

X, Y indicate red on two draws from three red and two blue balls.

Without replacement

$X \backslash Y$	0	1	$p_X(x)$
0	1/10	3/10	2/5
1	3/10	3/10	3/5
$p_Y(y)$	2/5	3/5	1

Independent draws with replacement

$X \backslash Y$	0	1	$p_X(x)$
0	4/25	6/25	2/5
1	6/25	9/25	3/5
$p_Y(y)$	2/5	3/5	1

Compare the probability of two reds.

Marginals alone do not determine the joint law.

Marginals and an event probability

$X \backslash Y$	0	1	2	$p_X(x)$
0	.10	.25	.35	
1	.15	.05	.10	
$p_Y(y)$				1

Find both marginals and $P(Y > X)$. Mark the selected cells.

Joint PMFs and marginals

1. Define the experiment and both variables.
2. Label the possible values and enter zero for impossible pairs.
3. Check every entry is nonnegative and the grand total is one.
4. Sum rows for p_X and columns for p_Y.
5. Check each marginal sums to one.

Three different objects

An outcome belongs to the sample space.
A joint cell records a pair of values (x, y).
A marginal law records one variable's values and probabilities.

Reading the joint table

Recall the two draws **without replacement**.

$X \backslash Y$	0	1	$p_X(x)$
0	1/10	3/10	2/5
1	3/10	3/10	3/5
$p_Y(y)$	2/5	3/5	1

Describe each event and find its probability.

$$P(X = 1, Y = 1), \qquad P(X = 1).$$

Now suppose the first draw was red. What is the probability the second is red?

The observed first draw restricts us to the row $X = 1$.

Deriving a conditional PMF

Fix x with $p_X(x) > 0$

$$p_{Y|X}(y \mid x) = P(Y = y \mid X = x) = \frac{p_{X,Y}(x, y)}{p_X(x)}.$$

This is event conditioning with $A = \{Y = y\}$ and $B = \{X = x\}$.

Show that it is a PMF. Its entries are nonnegative. Check their sum.

Select a row, then divide every entry by that row's probability.

If $p_X(x) = 0$, this elementary ratio does not define a conditional PMF.

Condition on the first draw

Select the row $X = 1$

$X \backslash Y$	0	1	$p_X(x)$
0	1/10	3/10	2/5
1	3/10	3/10	3/5
$p_Y(y)$	2/5	3/5	1

Normalize by $p_X(1) = 3/5$

y	0	1
$p_{Y\mid X}(y \mid 1)$		

Compare with the marginal: $p_Y(1) = 3/5$.

Here observing $X = 1$ changes the law of Y. Under independence it would not.

Reversing the conditioning

Use the asymmetric joint table from Wednesday. Select the conditioning event first.

Fix the row $X = 1$

y	0	1	2
$p_{X,Y}(1, y)$	.15	.05	.10

The row total is $p_X(1) = .30$.

$$P(Y = 2 \mid X = 1) = \frac{.10}{.30}.$$

Fix the column $Y = 2$

x	$p_{X,Y}(x, 2)$
0	.35
1	.10

The column total is $p_Y(2) = .45$.

$$P(X = 1 \mid Y = 2) = \frac{.10}{.45}.$$

The numerator is the same cell; the conditioning event determines the denominator.

Recovering joint and marginal probabilities

For $p_X(x) > 0$, multiply the conditional-PMF definition by its denominator:

$$p_{X,Y}(x, y) = p_X(x) p_{Y|X}(y \mid x).$$

Then sum over x to recover the marginal.

$$p_Y(y) = \sum_{x \in S_X} p_X(x) p_{Y|X}(y \mid x), \qquad S_X = \{x : p_X(x) > 0\}.$$

Without replacement: after blue, 3 of 4 remaining balls are red; after red, 2 of 4.

Average conditional probabilities with their event probabilities as weights.

Rows with $p_X(x) = 0$ contain only zero joint masses and contribute nothing.

Independence and factorization

Independence means $P(X \in A, Y \in B) = P(X \in A)\,P(Y \in B)$ for all value sets A, B.

Equivalent criterion for discrete variables

$$p_{X,Y}(x, y) = p_X(x)p_Y(y) \quad \text{for every } (x, y) \in S_X \times S_Y.$$

Here S_X, S_Y are the marginal supports. Test zero joint cells too.

Finite proof. Use singleton events for one direction. For the converse, sum over $A \times B$.

Conditional laws under independence

Equivalent conditional criterion

X and Y are independent exactly when, for every x with $p_X(x) > 0$,

$$p_{Y|X}(y \mid x) = p_Y(y) \quad \text{for every } y.$$

Derive both directions by division or multiplication by $p_X(x)$.

One unequal cell refutes independence. One agreeing cell does not prove it.

Independence requires positive joint mass at every pair in $S_X \times S_Y$. This condition alone is insufficient.

Joint $p_{X,Y}(x,y) = P(X = x, Y = y)$
Subscripts: variables. Inputs: values (x, y).

↓ Sum

Marginals: sum out one variable

$$p_X(x) = \sum_{v \in S_Y} p_{X,Y}(x, v)$$

Fix x; only v runs through all Y-values.

$$p_Y(y) = \sum_{u \in S_X} p_{X,Y}(u, y)$$

Fix y; only u runs through all X-values.

↓ Joint ÷ marginal

Conditionals: fix the given value

$$p_{Y|X}(y \mid a) = \frac{p_{X,Y}(a,y)}{p_X(a)} \quad \big(p_X(a) > 0\big)$$

Input y varies; given $X = a$ stays fixed.

$$p_{X|Y}(x \mid b) = \frac{p_{X,Y}(x,b)}{p_Y(b)} \quad \big(p_Y(b) > 0\big)$$

Input x varies; given $Y = b$ stays fixed.

X and Y are independent

Three equivalent criteria; any one suffices

Joint factors into marginals

$$p_{X,Y}(x,y) = p_X(x)p_Y(y)$$

for every $x \in S_X$ and $y \in S_Y$

⇕

Conditional of Y equals its marginal

$$p_{Y|X}(y \mid a) = p_Y(y)$$

for every $y \in S_Y$ and a with $p_X(a) > 0$

⇕

Conditional of X equals its marginal

$$p_{X|Y}(x \mid b) = p_X(x)$$

for every $x \in S_X$ and b with $p_Y(b) > 0$

Conditioning leaves the other law unchanged.

Recover the joint when $p_X(a) > 0$

$$p_{X,Y}(a,y) = p_X(a)\, p_{Y|X}(y \mid a) \overset{\text{independence}}{=} p_X(a)\, p_Y(y)$$

Replacement changes the joint law

Without replacement

$X \backslash Y$	0	1	$p_X(x)$
0	1/10	3/10	2/5
1	3/10	3/10	3/5
$p_Y(y)$	2/5	3/5	1

Independent draws with replacement

$X \backslash Y$	0	1	$p_X(x)$
0	4/25	6/25	2/5
1	6/25	9/25	3/5
$p_Y(y)$	2/5	3/5	1

Both tables have positive mass in every cell. Only the replacement table factors.

Does this table show independence?

$X \backslash Y$	0	1	2	$p_X(x)$
0	.12	.18	.30	
1	.08	.12	.20	
$p_Y(y)$				1

Compute the marginals and test every cell. Then find $P(Y \geq 1 \mid X = 0)$.

Joint, marginal, and conditional PMFs

Object	Formula	Operation
Joint	$p_{X,Y}(x, y) = P(X = x, Y = y)$	Select one cell
Marginal	$p_X(x) = \sum_y p_{X,Y}(x, y)$	Sum a row
Conditional	$p_{Y\|X}(y \mid x) = \dfrac{p_{X,Y}(x, y)}{p_X(x)}$	Normalize a row

The conditional formula requires $p_X(x) > 0$.

Independence

The joint equals the product of marginals for every pair of marginal values.
Equivalently, every positive-probability conditional row reproduces p_Y.

Optional: the second moment of a binomial count

Let $I_1, \ldots, I_n$ be independent Bern(p) variables, and $N = \sum_i I_i$.

$$N^2 = \sum_{i=1}^{n} I_i^2 + 2 \sum_{i<j} I_i I_j.$$

Use $I_i^2 = I_i$. For $i \neq j$, $I_i I_j$ is the indicator of both successes.

Independence is used for the pair probability, not for linearity of expectation.

Optional: binomial variance

For $N \sim \mathrm{Bin}(n, p)$, the indicator calculation gives

$$\mathbb{E}[N] = np, \qquad \mathbb{E}[N^2] = np + n(n-1)p^2.$$

Apply the variance identity and simplify.

Binomial moments

$$\mathbb{E}[N] = np, \qquad \mathrm{Var}(N) = np(1-p).$$

Pairwise independence suffices for this variance calculation. The binomial trial model assumes mutual independence. General variance-of-sums rules are taught later.

Optional: the mean minimizes squared error

Assume $\mathbb{E}[X^2] < \infty$ and write $\mu = \mathbb{E}[X]$. For any real constant c,

$$\mathbb{E}[(X - c)^2] = \mathrm{Var}(X) + (\mu - c)^2.$$

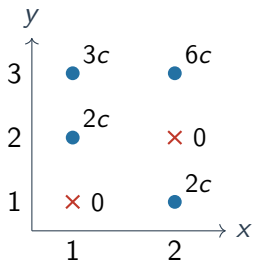
Derive it: write $X - c = (X - \mu) + (\mu - c)$, expand, and take expectation.

Variance is the smallest expected squared error from using a constant prediction.

Optional: support matters for independence

Let $S = \{(x, y) : x \in \{1, 2\},\ y \in \{1, 2, 3\},\ x \neq y\}$.

$$p_{X,Y}(x, y) = cxy\mathbb{I}_{\{(x,y)\in S\}}.$$



Normalize, find marginals, test $(1, 1)$**.**

The factor xy alone does not establish independence. Check the support too.

Reading selections

DeGroot and Schervish, *Probability and Statistics*, 4th ed.

- Variance: §4.1, p. 213 (discrete function-expectation proof); §4.3, pp. 226–229 (basic properties).
- Joint laws: §3.4, pp. 118–120; §3.5, pp. 131–132 (discrete marginals).
- Independence/conditioning: §3.5, pp. 134–136; §3.6, pp. 142–143 (discrete parts).

Harchol-Balter, *Introduction to Probability for Computing*, 2024 version.

- Moments/variance: p. 83 (definition); pp. 85–87 (variance and SD); p. 88 (Lemma 5.7 only).
- Joint PMFs/independence: §3.3, pp. 50–51; p. 53 (total-probability statement).
- Conditional PMFs: §4.3, pp. 67–68, stopping before conditional expectation.

Printed page numbers. Selections only; skip continuous arguments and later topics. Lectures are self-contained. Additional reference: Sheng Jiang, STAT 131 Notes (July 18, 2024), pp. 34, 38–43, 54–55, selected discrete statements.